Polynomial ergodic averages on arbitrary intervals

A literature-implied answer to Austin's first further question

Status: `literature_implied_answer` Updated 27 August 2026

Source. Tim Austin, *Norm convergence of continuous-time polynomial multiple ergodic averages*, arXiv:1103.0223. In Section 4, "Further questions", PDF page 26, Austin asks whether Theorem 1.2 remains valid over arbitrary real intervals whose lengths tend to infinity, independently of their distance from the origin. In the paper's notation, for a measurable measure-preserving $\mathbb{R}^D$-action τ, polynomial maps $p_i : \mathbb{R} \to \mathbb{R}^D$, and $f_i \in L^\infty(\mu)$, the question is whether

$$A_I(f_1, \ldots, f_k) := \frac{1}{|I|} \int_I \prod_{i=1}^{k} f_i \circ \tau^{p_i(t)} \, dt$$

converges in $L^2(\mu)$ whenever $|I| \to \infty$, with no restriction on the location of I.

Supporting theorem. Pavel Zorin-Kranich, *Norm convergence of multiple ergodic averages on amenable groups*, arXiv:1111.7292, Theorem 1.1(1), PDF page 2, proves the following. Let $\mathcal{G}$ be a locally compact amenable group, $\mathcal{T}$ a nilpotent group of measure-preserving Koopman operators, and $T_i : \mathcal{G} \to \mathcal{T}$ measurable polynomial maps. Then the associated multiple averages converge in L^2 along every left Folner net, and the limit does not depend on the Folner net. Section 4 of that paper defines polynomial maps by iterated discrete derivatives; its example following Theorem 4.5 explicitly includes conventional real polynomials composed with one-parameter subgroups.

Exact identification. Set $\mathcal{G} = (\mathbb{R}, +)$ and let $U_v f = f \circ \tau^v$. The image

$$\mathcal{T} = \{U_v : v \in \mathbb{R}^D\}$$

is abelian, hence nilpotent. Each $T_i(t) := U_{p_i(t)}$ is a measurable polynomial map in Zorin-Kranich's sense. Thus Theorem 1.1 applies to Austin's integrand (take the harmless factor $f_0 = 1$).

Now let I_n be arbitrary intervals with $|I_n| \to \infty$. For every compact $K \subset \mathbb{R}$,

$$\sup_{h \in K} \frac{|(h + I_n) \triangle I_n|}{|I_n|} \leq \frac{2 \sup_{h \in K} |h|}{|I_n|} \longrightarrow 0.$$

Hence (I_n) is a Folner sequence, regardless of how far its members lie from the origin. Zorin-Kranich's theorem therefore gives the requested L^2 convergence and shows that the limit is independent of the chosen moving interval sequence.

Conclusion. *Austin's arbitrary-interval question has an affirmative answer: the averages $A_I(f_1, \ldots, f_k)$ converge in $L^2(\mu)$ as $|I| \to \infty$, uniformly in the sense that every sequence of intervals with diverging lengths has the same limit.*

Additional scope. The same specialization with $\mathcal{G} = \mathbb{R}^r$ also answers Austin's Question 4.1 (PDF page 26): polynomial averages over $[0, R)^r$ converge, and in fact they converge along arbitrary Folner nets in $\mathbb{R}^r$. It also covers the final extension on PDF page 27 to actions of a connected nilpotent Lie group, provided "polynomial" is understood in the discrete-derivative (Leibman) sense used by the supporting theorem. The intervening Hardy-field question is outside this polynomial theorem and is not classified here.

Provenance and novelty status. This is not a new theorem produced by the run. Zorin-Kranich cites Austin's work but does not explicitly identify the Section 4 question as the target of Theorem 1.1; the relation above is a direct specialization found during this run. That is why the packet is classified as a literature-implied answer rather than an explicitly literature-answered problem. A bounded search by arXiv id, exact title, "arbitrary intervals", "uniform Cesaro", and "Folner" also found Bergelson–Leibman–Moreira's later Folner-sequence theorem for one-parameter flows; Zorin-Kranich's theorem supplies the decisive full $\mathbb{R}^D$-action scope.

Human review recommendation. High confidence. Verify only the standard functorial point that composing the $\mathbb{R}^D$-action with an ordinary vector polynomial gives a polynomial operator-valued map in Definition 4.3 of arXiv:1111.7292. The abelian case is also illustrated explicitly after its Theorem 4.5.

References

[1] T. Austin, *Norm convergence of continuous-time polynomial multiple ergodic averages*, Ergodic Theory Dynam. Systems 32 (2012), 361–382; arXiv:1103.0223.

[2] P. Zorin-Kranich, *Norm convergence of multiple ergodic averages on amenable groups*, J. Analyse Math. 130 (2016), 219–241; arXiv:1111.7292.

[3] V. Bergelson, A. Leibman, and C. G. Moreira, *From discrete- to continuous-time ergodic theorems*, Ergodic Theory Dynam. Systems 32 (2012), 383–426.